Customer Support Ticket Analysis

1 Introduction

This report presents an exploratory data analysis (EDA) and preliminary natural language processing (NLP) analysis of a large customer support ticket dataset. The goal is to understand the main types of issues raised by customers and identify key patterns that can inform further analysis.

2 Dataset Overview

The dataset contains 61,765 customer support tickets with 16 features, including textual fields (e.g., subject, body), categorical variables (e.g., type, priority, queue), and multiple tag annotations.

2.1 Missing Values

The dataset contains a significant number of missing values, particularly in tag fields and the `version` column. However, key variables such as `body`, `priority`, `queue`, and `language` are largely complete, making them suitable for analysis.

3 Exploratory Data Analysis

3.1 Ticket Type Distribution

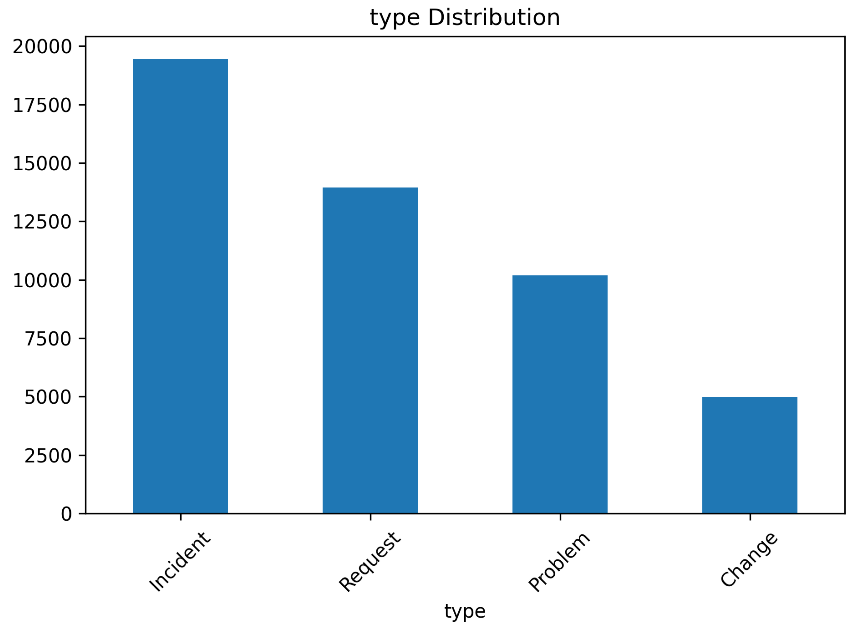

Figure 1: Distribution of Ticket Types

Most tickets are classified as **Incident**, followed by Request and Problem. This suggests that customers primarily report unexpected issues or disruptions rather than planned changes.

3.2 Priority Distribution

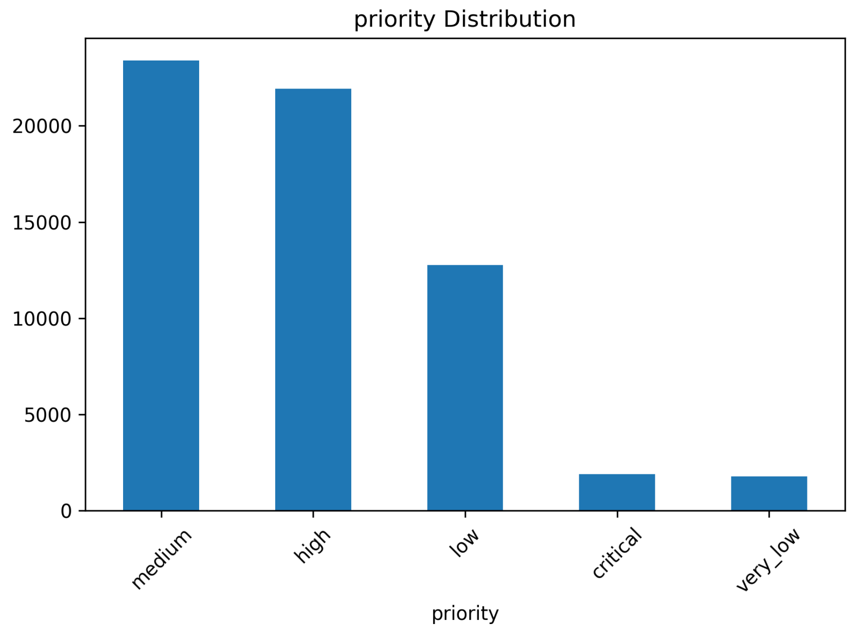

Figure 2: Distribution of Ticket Priority

The majority of tickets are of medium and high priority, indicating that many issues require timely attention. However, relatively few tickets are critical, suggesting that severe failures are less frequent.

3.3 Queue Distribution

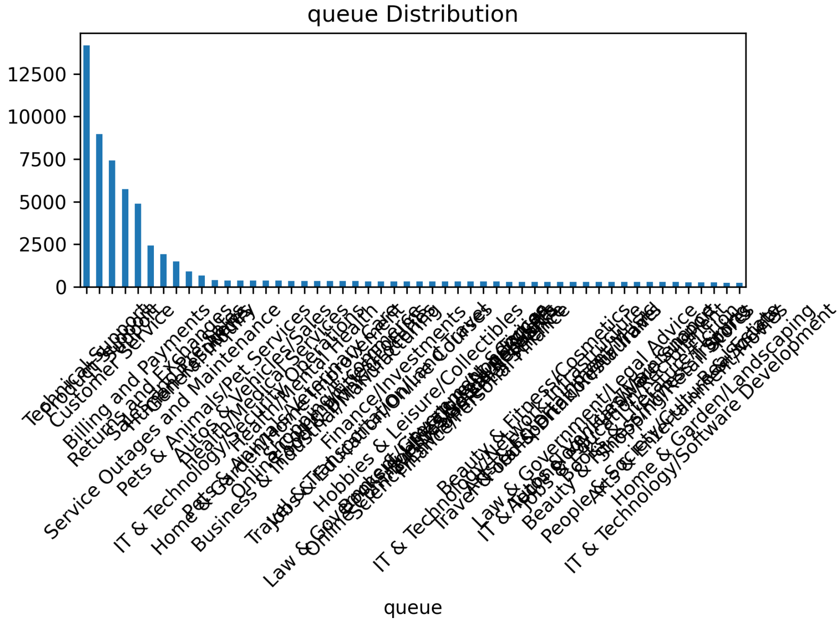

Figure 3: Top Queues

The Technical Support queue handles the largest number of tickets, followed by Product Support and Customer Service. This indicates that most issues are technical in nature.

3.4 Language Distribution

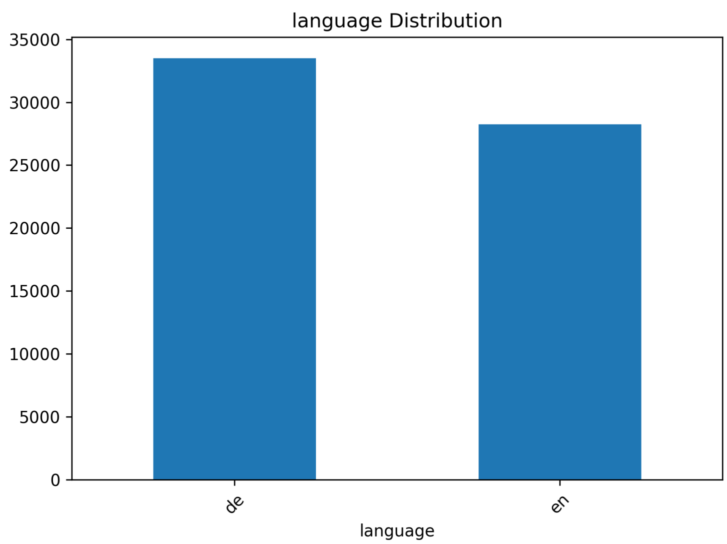

Figure 4: Language Distribution

The dataset is dominated by German-language tickets, with a substantial number of English tickets. This suggests a primarily German-speaking customer base with international users.

3.5 Tag Analysis

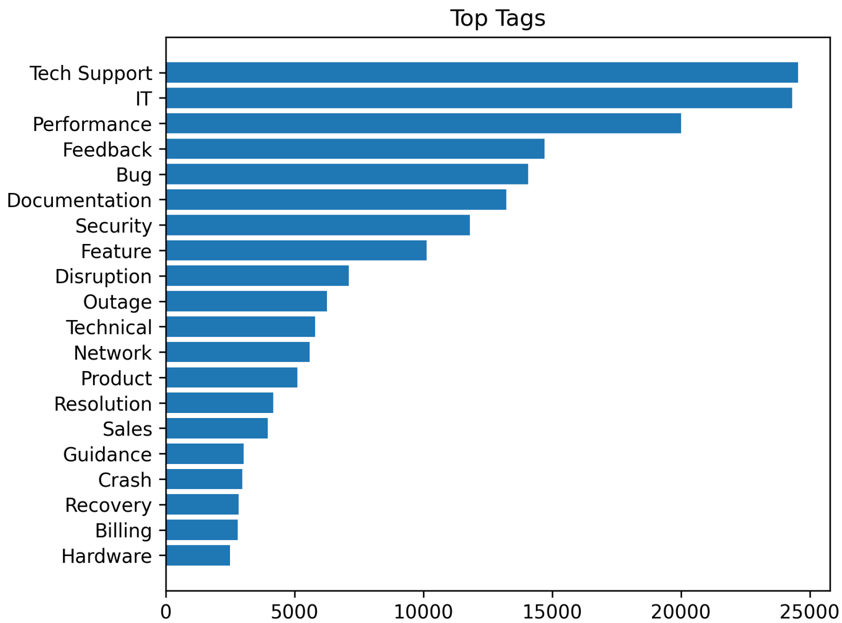

Figure 5: Top Tags

The most frequent tags include:

- Tech Support and IT
- Performance issues
- Bugs and software errors
- Security concerns

This indicates that customer issues are heavily concentrated around technical system performance and reliability.

3.6 Text Length Analysis

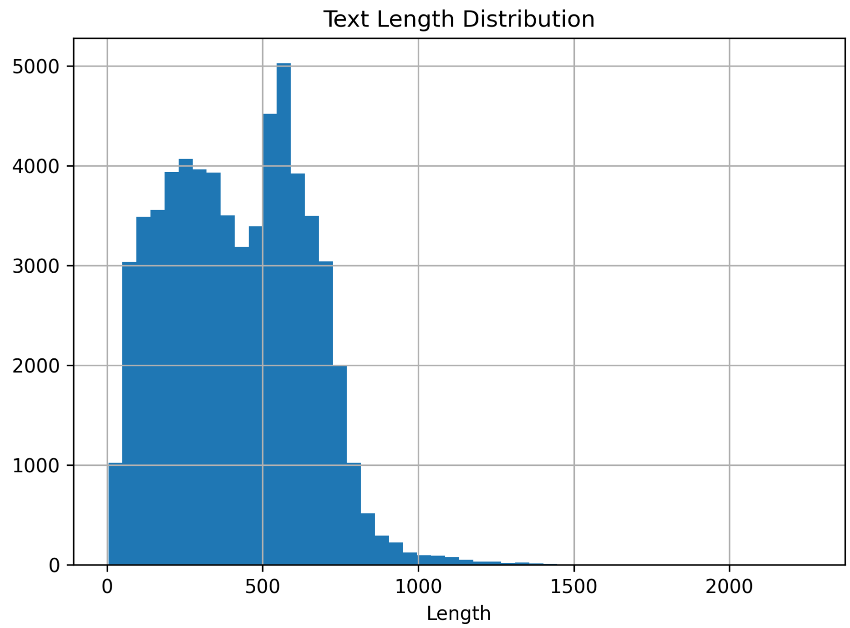

Figure 6: Text Length Distribution

The average ticket length is approximately 419 characters, indicating that customers generally provide moderately detailed descriptions. Some tickets are significantly longer, suggesting more complex issues.

3.7 Cross Analysis

Type vs Priority:

Incident tickets are more likely to be associated with higher priority levels, indicating that unexpected system issues tend to be more critical.

Language vs Type:

The distribution of ticket types is similar across languages, suggesting consistent issue patterns across regions.

4 NLP-Oriented Analysis

4.1 Word Frequency

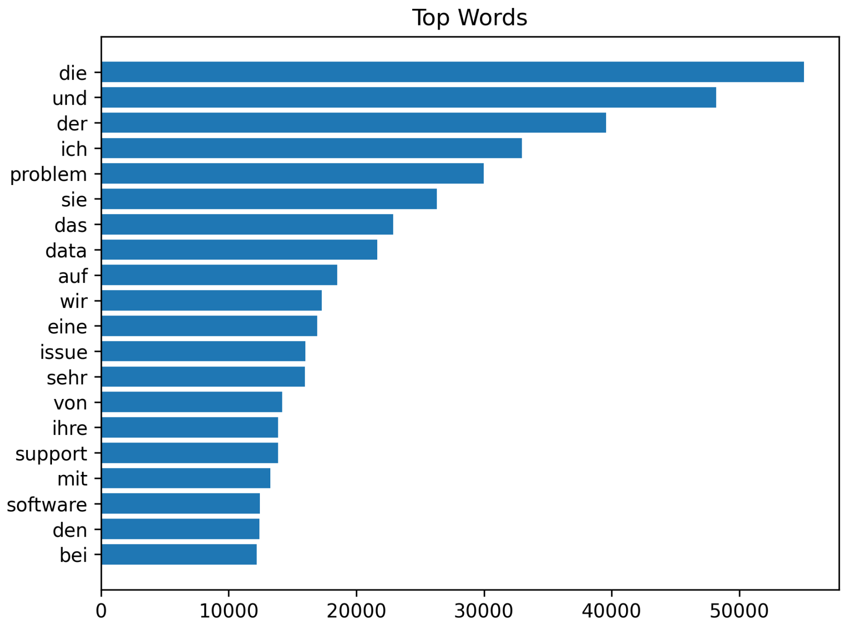

Figure 7: Top Words

The most frequent words include both German and English terms such as *problem*, *data*, *support*, and *software*. This reflects the multilingual nature of the dataset and highlights recurring technical issues.

4.2 Bigram Analysis

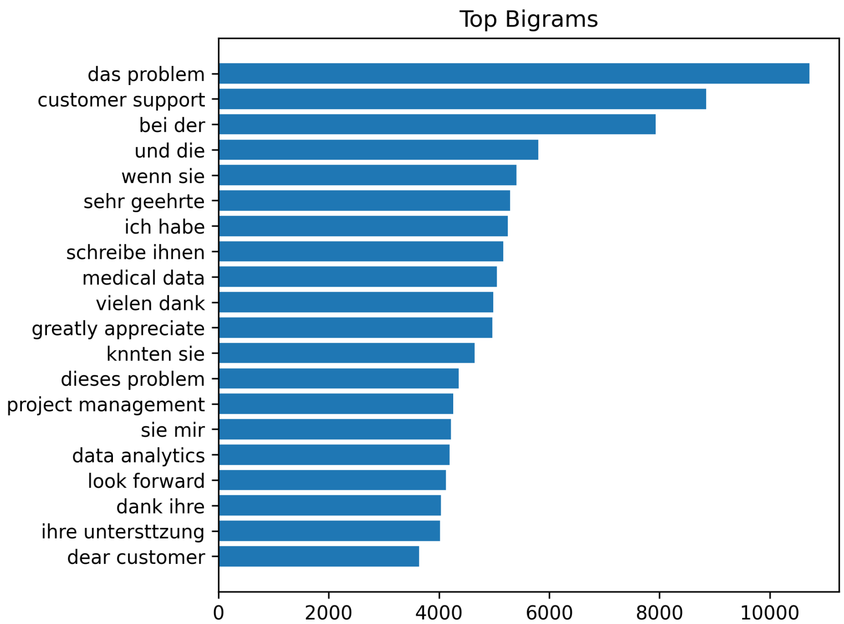

Figure 8: Top Bigrams

Common bigrams such as *customer support*, *data analytics*, and *das problem* reveal frequently discussed themes, including technical issues and customer service interactions.

4.3 Word Length Analysis

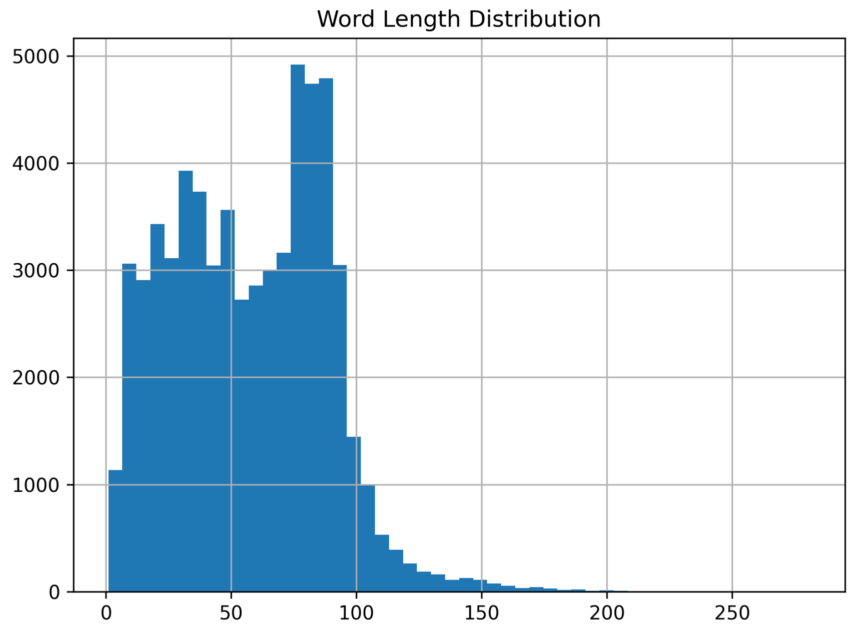

Figure 9: Word Length Distribution

The average ticket contains approximately 57 words, suggesting that users provide sufficient detail for meaningful text analysis.

4.4 Word Analysis by Ticket Type

Different ticket types exhibit distinct vocabulary patterns:

- Incident and Problem tickets emphasize terms such as *problem*, *issue*, and *software*
- Request tickets include words like *information*, *provide*, and *integration*
- Change tickets focus on *requests*, *tools*, and *customer needs*

This indicates that different ticket categories correspond to different types of user intent.

5 Conclusion

The analysis shows that customer support tickets are primarily driven by technical issues, particularly incidents related to system performance, bugs, and software reliability. Most tickets are of medium to high priority, indicating a significant operational workload.

The NLP analysis further reveals recurring themes such as technical problems, customer support interactions, and data-related issues. These findings suggest that improving system performance, enhancing technical support, and addressing common bugs could significantly reduce the volume of support tickets.